STUDY MATERIAL GRADE 7

PHYSICS

Detailed notes for every textbook section

39 self-contained sections: key ideas, accurate definitions, formulas, units and visual diagrams. The calculation sections also include worked examples based on the textbook exercises.

Based on: N. A. Zakirova, R. R. Ashirov and A. V. Filipov, Physics: Textbook for Grade 7, Astana, 2025.

English translation of the supplied Russian study notes. The content has been condensed for revision; diagrams are original vector illustrations.

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Each numbered section is self-contained. Worked solutions are included after calculation-based sections.

Based on Zakirova, Ashirov & Filipov (2025)

Chapter I. Physics: the Science of Nature

1

Physics: the Science of Nature

Physics studies mechanical, thermal, sound, light, electrical and magnetic phenomena.

Its task is to observe phenomena, explain their causes and discover the laws of nature.

Precise physical terms make it possible to describe bodies, substances and their properties unambiguously.

DEFINITIONS

Physics	the science that studies physical phenomena.
Physical phenomenon	a change in which one substance does not turn into another substance.
Physical body	any object that has shape and volume.
Substance and matter	a substance is a type of matter with specific properties; matter is everything that exists around us. Matter may exist as substance or as a field.

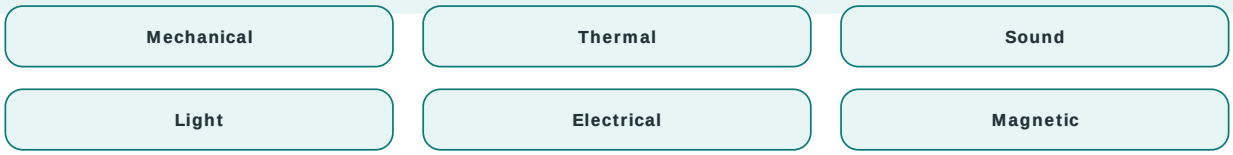
FORMULAS AND RELATIONS

No formulas in this section

The main point is to identify types of phenomena and use terms precisely.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE We see lightning before we hear thunder because light travels faster than sound.

Chapter I. Physics: the Science of Nature

2

Scientific Methods for Studying Nature

Scientific inquiry begins with purposeful observation.

A hypothesis explains an observed phenomenon, but it must be tested.

Repeated experiments lead to a pattern and then to a physical theory.

DEFINITIONS

Observation	purposeful perception of a phenomenon.
Hypothesis	an assumption that explains an observed phenomenon and requires proof.
Experiment	a method of studying a phenomenon under specially created and controlled conditions.
Physical theory	a system of scientific knowledge that explains physical phenomena and their relationships.

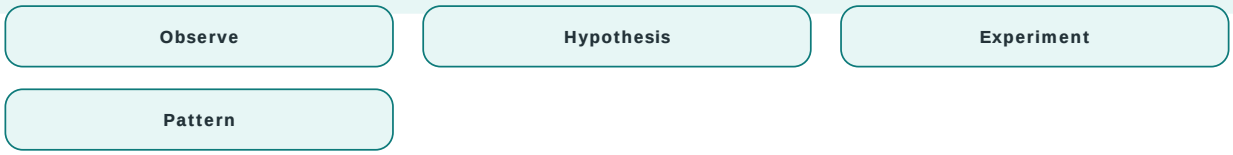
FORMULAS AND RELATIONS

observation hypothesis experiment pattern theory

This is the logical sequence of scientific inquiry.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE In an experiment, conditions are changed deliberately, results are measured, and the hypothesis is checked.

Chapter II. Physical Quantities and Measurement

3

The International System of Units (SI)

A physical quantity describes a measurable property of a body or a phenomenon.
Measuring means comparing a quantity with a homogeneous quantity chosen as a unit.
A common system of units lets everyone interpret results in the same way.

DEFINITIONS

Physical quantity	a quantitative characteristic of a property of a body or phenomenon.
Measurement	comparison of a quantity with its unit.
International System of Units (SI)	an agreed system of base and derived units of physical quantities.

FORMULAS AND RELATIONS

SI base units	length metre (m); mass kilogram (kg); time second (s); electric current ampere (A); temperature kelvin (K); amount of substance mole (mol); luminous intensity candela (cd).
Examples of derived units	speed m/s; volume m ; density kg/m ; force newton (N); pressure pascal (Pa).

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Before calculating, express every quantity in SI. Prefixes: kilo = 10 , centi = 10 , milli = 10 , micro = 10 .

Chapter II. Physical Quantities and Measurement

3 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 3

Problem. Find the volume of a cube with side 10 cm. Give the result in m , dm , cm and mm .

Given	SI	Solution
a = 10 cm V ?	a = 0.1 m	V = a V = (0.1 m) = 0.001 m 0.001 m = 1 dm = 1000 cm = 1 000 000 mm

Answer V = 0.001 m = 1 dm = 1000 cm = 1 000 000 mm .

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter II. Physical Quantities and Measurement

4

Scalar and Vector Physical Quantities

A scalar quantity needs only a numerical value and a unit.

A vector quantity also has a direction and is represented by an arrow.

To read a measuring instrument correctly, first determine the value of one scale division.

DEFINITIONS

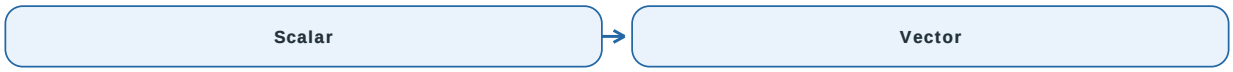
Scalar quantity	a quantity specified by a numerical value and unit only: mass, time, volume or distance travelled.
Vector quantity	a quantity with a magnitude and direction: force, displacement and displacement velocity.
Instrument scale	a reading panel marked with divisions and numbers.
Value of a division	the value of a quantity corresponding to the smallest division of a scale.

FORMULAS AND RELATIONS

c = (A A) / n	c is the value of a division; A and A are neighbouring labelled readings; n is the number of intervals between them.
A = A + n c	a reading equals the nearest labelled value A plus n divisions multiplied by c.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE The arrow of a vector shows its direction; its length, in the chosen scale, shows the magnitude.

Chapter II. Physical Quantities and Measurement

4 WORKED EXAMPLE

Worked Textbook Problem

Worked example in 4

Problem. A measuring cylinder has labelled marks 400 mL and 500 mL with 10 intervals between them. The liquid level is 6 divisions above 500 mL. Find the value of one division and the volume.

Given	SI	Solution
A = 400 mL A = 500 mL n = 10 N = 6 div. V ?	No conversion needed	c = (A A)/n c = (500 400)/10 = 10 mL V = 500 mL + 6 10 mL = 560 mL

Answer One division c = 10 mL; liquid volume V = 560 mL.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter II. Physical Quantities and Measurement

5

Accuracy of Measurements and Calculations. Scientific Notation

Every measurement is approximate: instruments, reading the scale and experimental conditions all affect it.

For a direct measurement, the reading uncertainty is usually taken as half of the scale division.

Scientific notation makes calculations with very large and very small numbers easier.

DEFINITIONS

Absolute error	the possible deviation of a measured value from the true value; it has the same unit as the quantity.
Reading error	an uncertainty caused by reading a scale; for one reading it is half a division.
Scientific notation	the form a 10 , where 1 a < 10 and n is an integer.

FORMULAS AND RELATIONS

A = Ainstr + Aread; Aread = c/2	A is absolute error; c is the value of one scale division.
A = Ameas A	the true value lies between Ameas A and Ameas + A.
(a 10)(b 10) = ab 10	when multiplying, add exponents; when dividing, subtract them. For addition, first make the powers of ten equal.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Example: 0.000000354 = 3.54 10 . Record a measured value to the same precision as its uncertainty.

Chapter II. Physical Quantities and Measurement

5 WORKED EXAMPLE

Worked Textbook Problem

Task 1 in 5

Problem. Write 1 250 000 000 and 0.000000125 in scientific notation.

Given	SI	Solution
a = 1 250 000 000 a = 0.000000125	No unit conversion	a = 1.25 10 a = 1.25 10

Answer 1 250 000 000 = 1.25 10 ; 0.000000125 = 1.25 10 .

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter III. Mechanical Motion

6

Mechanical Motion and Its Characteristics. Reference Frame

Mechanical motion is always considered relative to a chosen reference body.

An extended body can be treated as a material point when its size is negligible for the problem.

Trajectory, distance travelled and displacement must not be confused.

DEFINITIONS

Mechanical motion	a change of a body s position in space over time relative to other bodies.
Reference frame	a reference body, a coordinate system attached to it, and an instrument for measuring time.
Trajectory	the line described by a moving body relative to a reference body.
Distance and displacement	distance l is the length of the trajectory; displacement s is the directed segment from the initial position to the final position.

FORMULAS AND RELATIONS

s = x x	the displacement projection is the final coordinate minus the initial coordinate.
l |s|	the magnitude of displacement cannot exceed distance travelled; equality holds for straight motion without a change of direction.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Distance is a scalar, while displacement is a vector. On returning to the starting point, displacement is zero but distance is not.

Chapter III. Mechanical Motion

6 WORKED EXAMPLE

Worked Textbook Problem

Worked example 1 in 6

Problem. A material point moves on Ox from coordinate 2 m to coordinate 5 m. Find the displacement projection.

Given	SI	Solution
x = 2 m x = 5 m s ?	All values are in SI	s = x x s = 5 m 2 m s = 3 m

Answer s = +3 m. The plus sign indicates the Ox direction.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter III. Mechanical Motion

7

Relativity of Mechanical Motion

Rest and motion are not absolute: they depend on the reference body selected.

The same body may move relative to Earth and be at rest relative to another body.

Trajectory and speed also depend on the reference frame.

DEFINITIONS

Relativity of motion	the dependence of motion on the selected reference frame.
Relativity of trajectory	the same body can have different trajectories for different observers.
Relative velocity	a velocity that has meaning only relative to a chosen reference body.

FORMULAS AND RELATIONS

vrel = vbody vframe	vector relation for velocities; for one-dimensional motion, signs show direction.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE A passenger is at rest relative to a carriage but moves relative to Earth. A point on a wheel traces a circle relative to its axle and a cycloid relative to the road.

Chapter III. Mechanical Motion

8

Straight Uniform and Non-uniform Motion

The type of motion is determined by distances covered in any equal time intervals.
In uniform motion those distances are equal; in non-uniform motion they differ.
Uniform motion is a useful model; real bodies often move uniformly only over part of a journey.

DEFINITIONS

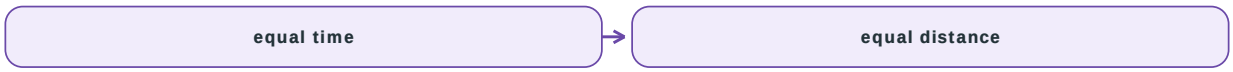
Uniform motion	motion in which a body covers equal distances in any equal time intervals.
Non-uniform motion	motion in which a body covers unequal distances in equal time intervals.
Straight-line motion	motion along a straight trajectory.

FORMULAS AND RELATIONS

Uniform: l = l = when t = t =	equal distances must correspond to any equal time intervals.
Non-uniform: l , l , differ for equal t	the body speeds up, slows down, or has different speeds on different sections.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Planets move non-uniformly along curved orbits: their speed is greater when they are closer to the Sun.

Chapter III. Mechanical Motion

9

Calculating Speed and Average Speed

Path speed describes how fast a body moves along its trajectory and is a scalar.

Displacement velocity is a vector; its direction is the direction of displacement.

Average speed must be found from total distance and total time, not by simply averaging speeds.

DEFINITIONS

Path speed	a quantity describing how fast a body moves along a trajectory.
Displacement velocity	a vector quantity describing the rate of displacement in a chosen direction.
Average speed	the speed of uniform motion that would cover the same total distance in the same total time.

FORMULAS AND RELATIONS

v = l/t	path speed; [v] = m/s. Hence l = vt and t = l/v.
v = s /t = (x x)/t	the projection of displacement velocity on the x-axis.
vavg = (l +l +)/(t +t +)	average speed over several sections.
x = x + v t	the equation of straight uniform motion.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE 1 m/s = 3.6 km/h; 1 km/h 0.28 m/s. On a closed trajectory, displacement and displacement velocity are zero.

Chapter III. Mechanical Motion

9 WORKED EXAMPLE

Worked Textbook Problem

Worked example in 9

Problem. A commuter train travels 300 km at 100 km/h and returns over the same route at 50 km/h. Find the average speed for the whole trip.

Given	SI	Solution
l = l = 300 km v = 100 km/h v = 50 km/h vavg ?	No conversion needed	t = l /v = 300/100 = 3 h t = l /v = 300/50 = 6 h vavg = (l +l)/(t +t) = 600/9 = 66.7 km/h

Answer vavg = 66.7 km/h.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter III. Mechanical Motion

10

Graphical Representation of Mechanical Motion

A graph shows how one quantity depends on another; both axes need quantities and units.
For uniform motion, v(t) is a horizontal line and l(t) is a straight line through the origin.
On l(t), speed is the slope; on v(t), distance is the area under the graph.

DEFINITIONS

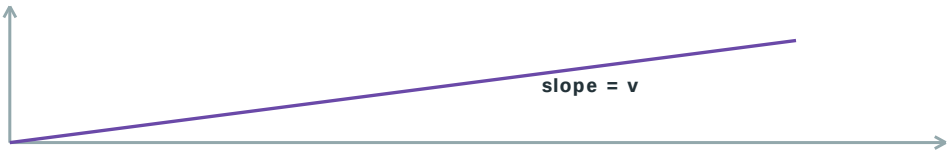
Graph of a constant quantity	a line parallel to the time axis: the quantity does not change.
Direct proportionality	a relationship y = kx; its graph is a straight line through the origin.
Slope	the ratio of the vertical change to the horizontal change on a graph.

FORMULAS AND RELATIONS

For uniform motion: l = vt	therefore l(t) is a straight line through the origin.
v = l/ t	speed is found from the slope of a distance time graph.
l = area under v(t)	for constant speed, l = vt is the area of a rectangle.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE A horizontal coordinate graph x(t) means that the body is at rest. A downward slope means a negative velocity projection.

Chapter III. Mechanical Motion

10 WORKED EXAMPLE

Worked Textbook Problem

Example from Fig. 36 in 10

Problem. On a uniform-motion speed graph, v = 3 m/s. Find the distance travelled in 4 s.

Given	SI	Solution
v = 3 m/s t = 4 s l ?	All values are in SI	l = vt l = 3 m/s 4 s l = 12 m

Answer l = 12 m. It is also the area of the rectangle under v(t).

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IV. Density

11

Mass and Measuring the Mass of Bodies

Mass characterizes the inertial and gravitational properties of a body.

The greater the mass, the harder it is to change a body s velocity and the stronger Earth s attraction is.

Mass is conserved and additive.

DEFINITIONS

Inertia	a property of bodies: when they interact, their state of rest or motion changes gradually, not instantaneously.
Mass	a scalar quantity characterizing a body s inertial and gravitational properties.
Additivity of mass	the mass of a body or system equals the sum of the masses of its parts.
Weighing	determining mass by comparison with known masses or by a body s action on scales.

FORMULAS AND RELATIONS

m = m + m +	the additivity of mass.
m /m = v / v	in interaction, changes in velocity are inversely proportional to masses.
1 t = 10 kg; 1 g = 10 kg; 1 mg = 10 kg	the SI unit of mass is the kilogram.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Beam balances compare masses with weights; spring balances use spring extension; electronic balances use sensors.

Chapter IV. Density

11 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 11

Problem. A body on a beam balance is balanced by weights of 20 g, 10 g, 2 g, 1 g and 500 mg. Find its mass in grams and kilograms.

Given	SI	Solution
m = 20 g m = 10 g m = 2 g m = 1 g m = 500 mg	500 mg = 0.5 g	m = m +m +m +m +m m = 20+10+2+1+0.5 m = 33.5 g = 0.0335 kg

Answer m = 33.5 g = 0.0335 kg.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IV. Density

12

Measuring the Volume of Regular and Irregular Bodies

The volume of a regular body is calculated from measured dimensions; this is an indirect measurement.

The volume of a small irregular body is found from the change in liquid level in a measuring cylinder.

In an overflow vessel, the volume of displaced water equals the volume of a fully submerged body.

DEFINITIONS

Volume	the space occupied by a body.
Direct measurement	determination of a quantity directly using an instrument.
Indirect measurement	calculation of a quantity using a formula and results of direct measurements.
Measuring cylinder	a graduated vessel for finding the volume of a liquid.

FORMULAS AND RELATIONS

$V_{cube} = a$; $V_{rectangular\ prism} = abc$	a, b and c are dimensions of the body.
$V_{cylinder} =$ r h; $V_{sphere} = 4$ r /3	r is radius and h is cylinder height.
$V_{body} = V$ V	for a body immersed in a measuring cylinder: final liquid volume minus initial volume.
1 m = 10 dm = 10 cm	1 dm = 1 L; 1 cm = 1 mL.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE The body must be completely covered by liquid and have no air bubbles attached to it.

Chapter IV. Density

12 WORKED EXAMPLE

Worked Textbook Problem

Task 2 in 12

Problem. Find the volume of the sphere on the Baiterek tower with diameter 22 m and compare it with the volume of the Nur-Alem pavilion with diameter 80 m.

Given	SI	Solution
d = 22 m d = 80 m V , V ?	r = 11 m r = 40 m	V = 4 r /3 V 4 3.14 11 /3 5.57 10 m V 4 3.14 40 /3 2.68 10 m V /V = (40/11) 48

Answer VBaiterek 5.57 10 m ; VNur-Alem 2.68 10 m . The second volume is about 48 times larger.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IV. Density

13

Density of a Substance and Density Calculations

Density shows what mass of a substance corresponds to unit volume.

For one substance under the same conditions, density is constant; different substances have different densities.

Knowing density allows mass or volume to be calculated.

DEFINITIONS

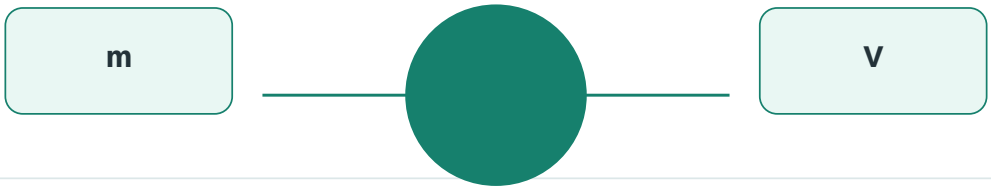
Density	a physical quantity equal to the ratio of a substance s mass to its volume.
Homogeneous body	a body whose density is the same in all parts.
Specific volume	the volume per unit mass; a quantity inverse to density.

FORMULAS AND RELATIONS

= m/V	is density, m is mass and V is volume.
m = V; V = m/	rearranged formulas.
[ ] = kg/m ; 1 g/cm = 1000 kg/m	the SI unit is kilogram per cubic metre.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE To find density experimentally, measure mass with scales, measure volume by shape or displacement, then divide m by V.

Chapter IV. Density

13 WORKED EXAMPLE

Worked Textbook Problem

Worked example in 13

Problem. What is the mass of a uranium sphere 10 cm in diameter? Density of uranium is 19 000 kg/m .

Given	SI	Solution
d = 10 cm = 19 000 kg/m m ?	d = 0.1 m r = 0.05 m	V = 4 r /3 V 4 3.14 (0.05) /3 0.00052 m m = V 19 000 0.00052 = 9.9 kg

Answer m 9.9 kg.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter V. Interaction of Bodies

14

The Phenomenon of Inertia

A body cannot start moving, stop, or change velocity by itself.
Without the action of other bodies, velocity is conserved: motion remains uniform and straight.
On Earth, friction and air resistance make pure inertia difficult to observe.

DEFINITIONS

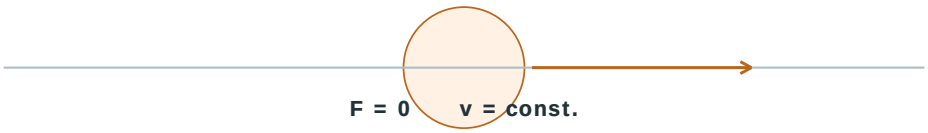
Inertia	the phenomenon of preserving a body s velocity when no other bodies act on it.
Inertia of rest	preservation of rest until interaction changes the state.
Inertia of motion	preservation of velocity by a moving body until interaction changes it.

FORMULAS AND RELATIONS

F = 0 v = const	if forces are absent or balanced, a body is at rest or moves uniformly in a straight line.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE When a bus brakes sharply, a passenger continues moving forward by inertia; a seat belt reduces the dangerous consequence.

Chapter V. Interaction of Bodies

15

Force

Force characterizes the interaction of bodies and can change velocity or produce deformation.

Force is a vector: magnitude, direction and point of application all matter.

Force is measured with a dynamometer whose operation is based on spring deformation.

DEFINITIONS

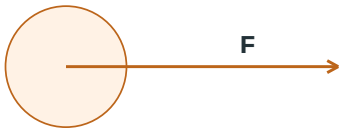
Force	a physical quantity characterizing interaction between bodies.
Force vector	a directed segment: its beginning is the point of application, arrow is direction, and scaled length is magnitude.
Dynamometer	an instrument for measuring force.

FORMULAS AND RELATIONS

[F] = 1 N	one newton is the force that changes the velocity of a 1 kg body by 1 m/s in 1 s.
1 kN = 10 N; 1 mN = 10 N; 1 MN = 10 N	multiples and submultiples of force.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE A vector force is written F. The symbol F without an arrow usually means only the magnitude.

Chapter V. Interaction of Bodies

15 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 15

Problem. Draw a 100 N force directed to the left using the scale 1 cm = 20 N.

Given	SI	Solution
F = 100 N 1 cm 20 N l ?	No conversion needed	l = F/(20 N/cm) l = 100/(20) = 5 cm

Answer Draw a vector 5 cm long pointing to the left.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter V. Interaction of Bodies

16

Gravitation, Gravitational Force and Weight

All bodies attract one another; this interaction is called universal gravitation.

Gravitational force is the force with which Earth attracts a body; it points toward Earth s centre.

Weight and gravitational force are different forces because their points of application differ.

DEFINITIONS

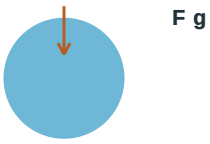
Gravitational force	the force with which Earth attracts a body.
Weight	the force with which a body, because of gravity, acts on a support or suspension.
Weightlessness	a state in which a body does not act on a support or suspension, although it still has mass and is acted on by gravity.
Gravitational field	the region of space in which Earth s attraction acts.

FORMULAS AND RELATIONS

Fg = mg	m is mass; near Earth s surface g 9.8 N/kg.
P = Fg = mg	for a body on a stationary horizontal support or hanging from a stationary vertical suspension.
[P] = [Fg] = N	weight is a force, so it is not measured in kilograms.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE At greater altitude Earth s attraction is smaller. A freely falling body is in weightlessness.

Chapter V. Interaction of Bodies

16 WORKED EXAMPLE

Worked Textbook Problem

Exercise 1 in 16

Problem. Find the gravitational force acting on 20 L of water.

Given	SI	Solution
V = 20 L = 1000 kg/m g = 10 N/kg F₫g• ?	V = 20 0.001 m V = 0.02 m	m = V m = 1000 0.02 = 20 kg F₫g• = mg F₫g• = 20 10 = 200 N

Answer Fg = 200 N.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter V. Interaction of Bodies

17

Deformation

When bodies interact, their shape or dimensions can change; this is deformation.
Elastic deformation disappears when the force is removed, while plastic deformation remains.
The type of deformation depends on the direction and point of application of the force.

DEFINITIONS

Deformation	a change in the shape and/or dimensions of a body.
Elastic deformation	a deformation after which the body completely restores its shape and size.
Plastic deformation	a deformation after which the body retains its new shape or size.
Types of deformation	tension, compression, shear, bending and torsion.

FORMULAS AND RELATIONS

external force deformation elastic force	an elastic body produces a restoring force when deformed.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE A spring and a rubber ball behave elastically; plasticine, wax and clay show plastic deformation.

Chapter V. Interaction of Bodies

18

Elastic Force and Hooke s Law

Elastic force appears when a body is deformed and is directed against the deformation.

Within the elastic limit, a spring s extension is directly proportional to the applied force.

A stiffer spring produces a greater elastic force for the same extension.

DEFINITIONS

Elastic force	the force arising during deformation that tends to restore a body s original shape and size.
Extension	the change in length: deformed length minus original length.
Spring constant	the coefficient of proportionality between elastic force and extension.

FORMULAS AND RELATIONS

Fel = k l	Hooke s law; k is spring constant and l is extension.
l = l l ; k = Fel/ l	l is the original length of the spring.
[k] = N/m	Hooke s law applies only to elastic deformation, before the elastic limit is exceeded.

VISUAL DIAGRAM

VISUAL DIAGRAM



F = k l

NOTE For a stationary load hanging on a spring, the elastic force equals the gravitational force in magnitude.

Chapter V. Interaction of Bodies

18 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 18

Problem. A spring is compressed by 8 mm under a force of 320 N. How much will it compress under a load of 1.6 kN?

Given	SI	Solution
F = 320 N l = 8 mm F = 1.6 kN l ?	F = 1600 N	F / l = F / l l = l F /F l = 8 1600/320 = 40 mm

Answer l = 40 mm.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter V. Interaction of Bodies

19

Friction Force. Friction in Engineering

Friction appears when bodies touch; it opposes relative motion or its beginning.

There are static friction, sliding friction and rolling friction.

Friction may be useful (walking, braking) or harmful (wear, heating and energy loss).

DEFINITIONS

Static friction	a force that prevents the start of relative motion between contacting bodies.
Sliding friction	a force during sliding of one body over another; it points opposite the motion.
Rolling friction	a force during rolling; it is usually less than sliding friction.
Coefficient of friction	a dimensionless number depending on materials and the condition of contacting surfaces.

FORMULAS AND RELATIONS

Ffr = N	for sliding friction: N is the normal reaction force.
Fstatic Fstatic,max	static friction adjusts to the applied force up to a maximum value.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE To reduce friction, use lubrication, polishing and bearings. To increase it, use roughness, rubber or studs.

Chapter V. Interaction of Bodies

19 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 19

Problem. A 2 kg wooden block is pulled uniformly along a wooden board. Find friction if , = 0.25.

Given	SI	Solution
m = 2 kg , = 0.25 g = 10 N/kg Ffr ?	All values are in SI	N = mg = 2 10 = 20 N Ffr = ,N Ffr = 0.25 20 = 5 N

Answer Ffr = 5 N.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter V. Interaction of Bodies

20

Adding Forces Acting along One Straight Line

Several forces can be replaced by one resultant force with the same effect on a body.

Forces in the same direction are added.

For opposite forces, find the difference and use the direction of the larger force.

DEFINITIONS

Resultant force	a force that produces the same effect as several forces acting at the same time.
Balanced forces	forces equal in magnitude and opposite in direction; their resultant is zero.
Tension force	the force with which a stretched string acts on a body.

FORMULAS AND RELATIONS

R = F + F +	when collinear forces act in the same direction.
R = |F F |	when forces have opposite directions; R points in the direction of the larger force.
R = 0	for equal opposite forces: a body is at rest or moves uniformly in a straight line.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE For n equally loaded vertical strings holding a load of weight P, the tension in each string is T = P/n.

Chapter V. Interaction of Bodies

20 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 20

Problem. A car is acted on by a tractive force of 1.25 kN, friction of 600 N and air resistance of 450 N. Find the resultant.

Given	SI	Solution
F = 1.25 kN F = 600 N F = 450 N R ?	F = 1250 N	R = F F F R = 1250 600 450 R = 200 N

Answer R = 200 N, directed along the car s motion.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

21

Molecular Structure of Solids, Liquids and Gases

All substances consist of tiny particles atoms and molecules with spaces between them.
Particles move continuously and randomly; diffusion demonstrates this.
Particles interact by attraction and repulsion; this explains the properties of states of matter.

DEFINITIONS

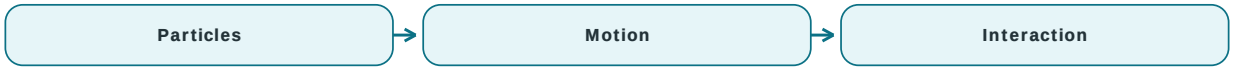
Atom	the smallest particle of a chemical element.
Molecule	the smallest particle of a substance that retains that substance s properties.
Diffusion	mutual penetration of particles of one substance between particles of another.
State of matter	a state with characteristic properties: solid, liquid or gaseous.

FORMULAS AND RELATIONS

Three statements of molecular-kinetic theory	1) substances consist of particles; 2) particles move continuously; 3) particles interact.
solid liquid gas	particle bonds weaken and freedom of motion increases; shape and volume are retained less and less.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Solids keep shape and volume; liquids keep volume but take the vessel s shape; gases have neither fixed shape nor volume and are easily compressed.

Chapter VI. Pressure

22

Pressure of Solids

The effect of a force on a surface depends both on the force and on the contact area.

The pressure force is always perpendicular to the surface.

Pressure increases when force grows or the support area decreases.

DEFINITIONS

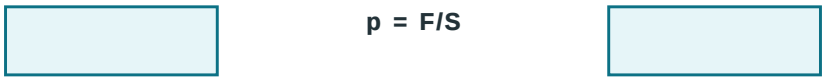
Pressure force	a force acting perpendicular to a surface.
Pressure	a quantity equal to force perpendicular to a surface divided by the area of that surface.
Support area	the area of contact between a body and its support.

FORMULAS AND RELATIONS

p = F/S	p is pressure, F is pressure force and S is support area.
F = pS; S = F/p	rearranged formulas.
1 Pa = 1 N/m	the SI unit of pressure is the pascal.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Skis and caterpillar tracks increase support area, reducing pressure. A sharp blade or needle decreases area, increasing pressure.

Chapter VI. Pressure

22 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 22

Problem. A roller of mass 6000 kg has a support area of 2000 cm . What pressure does it exert?

Given	SI	Solution
m = 6000 kg S = 2000 cm g = 10 N/kg p ?	S = 2000 10 ƒ m S = 0.2 m	F = mg = 6000 10 = 60 000 N p = F/S = 60 000/0.2 p = 300 000 Pa

Answer p = 3 10 Pa = 300 kPa.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

23

Pressure in Liquids and Gases. Pascal s Law

A liquid transmits pressure in all directions because it flows.

Liquid pressure increases with depth and with liquid density.

Gas pressure is caused by moving molecules striking the vessel walls; compression or heating raises it.

DEFINITIONS

Hydrostatic pressure	the pressure due to the weight of a liquid; it depends on density and depth.
Pascal s law	pressure applied to a liquid or gas is transmitted unchanged to every point in its volume.
Additional pressure	a pressure change produced by an external force above the original pressure.

FORMULAS AND RELATIONS

p = gh	pressure in a liquid at depth h; is density and g 9.8 N/kg.
F = pS	pressure force on a surface of area S.
pavg = (ptop + pbottom)/2	average pressure used to calculate force on a vertical side wall.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE In a gas, pressure rises when density or temperature rises. Unlike solids, liquids and gases transmit added pressure in all directions.

Chapter VI. Pressure

23 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 23

Problem. Find the pressure on the bottom from a 10 cm column of liquid with density 1800 kg/m .

Given	SI	Solution
= 1800 kg/m h = 10 cm g = 10 N/kg p ?	h = 0.1 m	p = gh p = 1800 10 0.1 p = 1800 Pa

Answer p = 1800 Pa = 1.8 kPa.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

24

Communicating Vessels

Communicating vessels are connected so that a liquid can flow between them.

With one homogeneous liquid, free surfaces settle at the same level if air pressure above them is the same.

For immiscible liquids, the greater density corresponds to the smaller column height.

DEFINITIONS

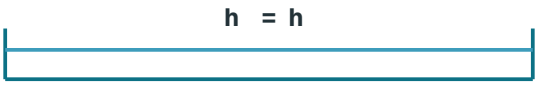
Communicating vessels	two or more vessels connected to one another.
Free surface of a liquid	the liquid surface in contact with a gas.
Arm of a communicating vessel	one branch of a communicating-vessel system.

FORMULAS AND RELATIONS

$p = p$; $gh = gh$	at equilibrium, pressure at the same horizontal level is equal.
$h = h$	for immiscible liquids: the greater the density, the smaller the height.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Applications include sight glasses, water supply, hydraulic systems, locks and artesian wells.

Chapter VI. Pressure

24 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 24

Problem. A lock is 10 m wide and filled with water to a depth of 5 m. With what force does water press on its gate?

Given	SI	Solution
b = 10 m H = 5 m = 1000 kg/m g = 10 N/kg	S = bH = 50 m h„= H/2 = 2.5 m	p„= gh„ p„= 1000 10 2.5 = 25 000 Pa F = p„S = 25 000 50 = 1 250 000 N

Answer F = 1.25 10 N = 1.25 MN.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

25

Hydraulic Machine

A hydraulic machine has two cylinders with pistons of different areas, connected by a tube filled with liquid.

By Pascal s law, the pressure on the two pistons is the same.

A large piston gives a gain in force but a loss in displacement.

DEFINITIONS

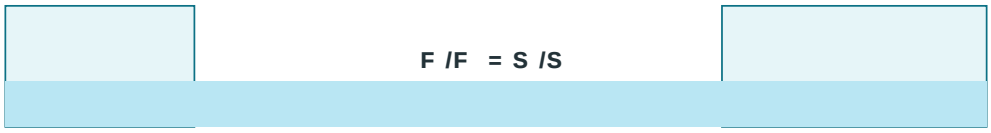
Hydraulic machine	a device whose action is based on the laws of motion and equilibrium of liquids.
Hydraulic press	a machine for treating materials by pressure: pressing, forging, stamping or briquetting.
Hydraulic jack	a device for lifting heavy loads by transmitting pressure through a liquid.

FORMULAS AND RELATIONS

p = p ; F /S = F /S	pressures on the small and large pistons are equal.
F /F = S /S	the force gain equals the area ratio of the large piston to the small piston.
F h = F h	in an ideal hydraulic machine, forces are inversely proportional to piston displacements.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE In a real machine, friction causes energy losses, so the real gain is less than the ideal one.

Chapter VI. Pressure

25 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 25

Problem. The large piston of a hydraulic press has area 180 cm and acts on oil with force 18 kN. The small piston has area 4 cm . Find its force.

Given	SI	Solution
S = 180 cm F = 18 kN S = 4 cm F ?	F = 18 000 N	F /S = F /S F = F S /S F = 18 000 4/180 = 400 N

Answer F = 400 N.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

26

Atmospheric Pressure and Its Measurement

The atmosphere is Earth s air envelope; its weight creates atmospheric pressure.

With altitude, air density and atmospheric pressure decrease.

Atmospheric pressure is measured by a mercury barometer or an aneroid barometer.

DEFINITIONS

Atmosphere	Earth s air envelope.
Atmospheric pressure	pressure exerted by the atmosphere on Earth s surface and bodies in the air envelope.
Barometer	an instrument for measuring atmospheric pressure.
Aneroid barometer	a liquid-free barometer that measures the elastic deformation of an evacuated metal capsule.

FORMULAS AND RELATIONS

patm = Hg g h	in Torricelli s experiment, atmospheric pressure balances a column of mercury.
1 mmHg = 133.3 Pa	relation between a non-SI unit and SI pressure unit.
pnormal • 760 mmHg • 101 300 Pa	normal atmospheric pressure at sea level.

VISUAL DIAGRAM

VISUAL DIAGRAM



h • 760 mm

NOTE For small altitude changes, pressure decreases by about 1 mmHg for every 12 m of ascent.

Chapter VI. Pressure

26 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 26

Problem. Find the mass of air in a room 4 … 5 … 3 m under normal conditions. Air density is 1.29 kg/m .

Given	SI	Solution
a = 4 m b = 5 m h = 3 m = 1.29 kg/m m ?	All values are in SI	V = abh = 4 5 3 = 60 m m = V = 1.29 60 = 77.4 kg

Answer m = 77.4 kg.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

27

Manometers and Pumps

A manometer measures pressure above or below atmospheric pressure.

A metal manometer uses deformation of a curved tube; a liquid manometer uses the difference in liquid levels in a U-tube.

Piston pumps use pressure differences and valves to move liquids.

DEFINITIONS

Manometer	an instrument for measuring pressure greater or less than atmospheric pressure.
Gauge pressure	the difference between gas pressure in a vessel and atmospheric pressure.
Liquid manometer	a U-shaped tube containing a liquid of known density.
Compressor	a device for compressing gas or air.

FORMULAS AND RELATIONS

$p = gh$	pressure difference from height difference h in a liquid manometer.
$p_{gas} = p_{atm} + p$	when gas pressure is greater than atmospheric; if smaller, $p_{gas} = p_{atm} - p$.
$1\ N/cm = 10,\ Pa$	conversion of a technical pressure unit.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE An ordinary piston pump relying on atmospheric pressure cannot lift water higher than about 10.3 m; an air chamber can create higher pressure.

Chapter VI. Pressure

27 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 27

Problem. Gas pressure is 8 N/cm . Convert it to pascals, hectopascals and kilopascals.

Given	SI	Solution
p = 8 N/cm	1 N/cm = 10ƒ Pa	p = 8 10ƒ Pa p = 80 000 Pa p = 800 hPa = 80 kPa

Answer p = 80 000 Pa = 800 hPa = 80 kPa.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

28

Buoyant Force

Liquid pressure increases with depth, so the force on the lower face of a submerged body is greater than the force on its upper face.

Their resultant is upward and is called buoyant force or Archimedes force.

Archimedes force depends on the density of the medium and the volume of the submerged part.

DEFINITIONS

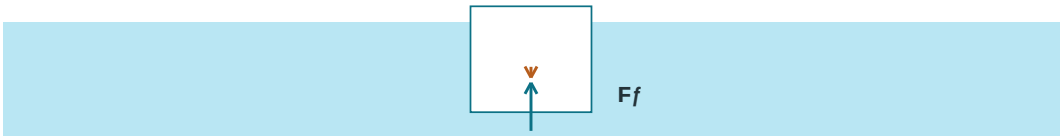
Buoyant force	the upward resultant of pressure forces of a liquid or gas on an immersed body.
Archimedes force	a buoyant force equal to the weight of the liquid or gas displaced by the immersed body.
Hydrostatic weighing	determining properties of a body from the change in its apparent weight when immersed.

FORMULAS AND RELATIONS

FA = liq g Vsub	Archimedes principle; Vsub is the volume of the submerged part.
Pliq = Pair FA	apparent weight in a liquid is smaller by Archimedes force.
body = liq Pair/(Pair Pliq)	formula for finding the density of a dense body by hydrostatic weighing.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Bodies of equal volume in the same liquid experience equal buoyant force regardless of their material. A denser liquid produces a larger force.

Chapter VI. Pressure

28 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 28

Problem. The submerged part of a patrol boat has volume 220 m . Find the buoyant force acting on it in water.

Given	SI	Solution
V = 220 m = 1000 kg/m g = 10 N/kg F¢A• ?	All values are in SI	F¢A• = gV F¢A• = 1000 10 220 F¢A• = 2 200 000 N

Answer FA = 2.2 10 N = 2.2 MN.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VI. Pressure

29

Conditions for Floating Bodies

A body in a liquid is acted on by gravitational force downward and Archimedes force upward.

Comparing these forces, and therefore densities, shows whether the body sinks, rises or floats within the liquid.

A body floating at the surface submerges until the forces balance.

DEFINITIONS

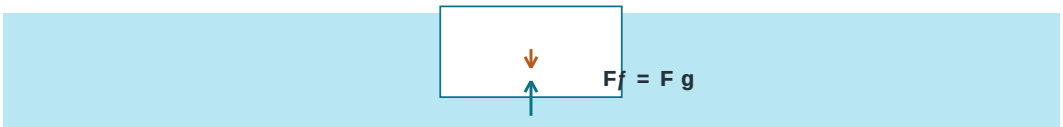
Floating within a liquid	equilibrium of a fully immersed body when gravitational and buoyant forces are equal.
Rising	upward movement when buoyant force is greater than gravitational force for the fully immersed body.
Hydrometer	an instrument that uses immersion depth to measure liquid density.
Centre of the submerged part	the point of application of Archimedes force.

FORMULAS AND RELATIONS

Fg = FA „ body = liq	the body floats within the liquid.
Fg > FA „ body > liq	the body sinks.
Fg < FA „ body < liq	the body rises; at the surface Fg = FA.
Vsub/Vbody = body/ liq	for a body floating on the surface.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Immiscible liquids form layers: denser liquid below and less dense liquid above.

Chapter VI. Pressure

29 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 29

Problem. A log floats in water with 2/5 of its volume submerged. Find the density of the wood.

Given	SI	Solution
V /V = 2/5 = 1000 kg/m ?	All values are in SI	V /V = / = V /V = 1000 2/5 = 400 kg/m

Answer wood = 400 kg/m .

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VII. Work and Power

30

Mechanical Work and Power

Mechanical work is done when a force acts on a body and moves it.

Work is positive when force and motion have the same direction, and negative when directions are opposite.

Power tells how quickly work is done.

DEFINITIONS

Mechanical work	a quantity characterizing the result of a force acting through a displacement.
Power	a quantity characterizing the rate at which work is done.
Tractive force	the force created by an engine and directed along a vehicle s motion.

FORMULAS AND RELATIONS

A = Fl	work by a force along the displacement; [A] = Ј = N m.
Afr = Ffrl	friction does negative work.
N = A/t	power; [N] = W = Ј/s.
N = Fv	relation of engine power, tractive force and speed in uniform motion.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE For the same power, increasing tractive force requires a lower speed; this is why vehicles slow down when climbing.

Chapter VII. Work and Power

30 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 30

Problem. A force does 30 kJ of work while moving a body through 7.5 m. Find the force.

Given	SI	Solution
A = 30 kJ l = 7.5 m F ?	A = 30 000 J	A = Fl F = A/l F = 30 000/7.5 = 4000 N

Answer F = 4000 N = 4 kN.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VIII. Energy

31

Kinetic Energy and Potential Energy

Energy characterizes a body s ability to do mechanical work; it is measured in joules.
Kinetic energy belongs to a moving body and depends on its mass and the square of speed.
Potential energy is interaction energy: of an elevated body with Earth or of a deformed elastic body.

DEFINITIONS

Mechanical energy	energy of mechanical motion and interaction of bodies or their parts.
Kinetic energy	energy of a moving body.
Potential energy	energy of interaction of bodies or parts of one body.
Work	a measure of the change in a body s energy.

FORMULAS AND RELATIONS

Ek = mv /2	kinetic energy.
Ep = mgh	potential energy of a body raised above a chosen level.
Eel = k(l) /2	potential energy of an elastically deformed spring.
A = Ek = Ek Ek	work equals the change in kinetic energy.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE If speed is tripled, kinetic energy becomes nine times greater. Doubling height doubles potential energy mgh.

Chapter VIII. Energy

31 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 31

Problem. A car of mass 1000 kg slows from 20 m/s to 10 m/s. Find the work done by the resultant force.

Given	SI	Solution
m = 1000 kg v = 20 m/s v = 10 m/s A ?	All values are in SI	A = Ek = m(v v)/2 A = 1000 (10 20)/2 A = 500 (100 400) A = 150 000 J

Answer A = 150 kJ. The negative sign shows that the resultant reduces speed.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter VIII. Energy

32

Transformation and Conservation of Energy

A body may have kinetic and potential energy at the same time.

During motion, energy of one type can transform into energy of another type.

In a closed system without significant friction, total mechanical energy is conserved.

DEFINITIONS

Total mechanical energy	the sum of a body s kinetic and potential energies.
Closed system	a system of bodies that interact with one another and practically do not interact with surroundings.
Conservation of mechanical energy	in a closed system where gravitational or elastic forces act, total mechanical energy remains constant.

FORMULAS AND RELATIONS

E = Ek + Ep	total mechanical energy.
E = E	law of conservation of total mechanical energy.
mgh + mv /2 = mgh + mv /2	for motion under gravity with negligible air resistance.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE With friction, part of mechanical energy turns into internal energy (heating) and sound, so mechanical energy decreases.

Chapter VIII. Energy

32 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 32

Problem. A 3 kg body falls freely from height 5 m. Find its potential and kinetic energies at height 2 m. Ignore air resistance.

Given	SI	Solution
m = 3 kg h = 5 m h = 2 m g = 10 N/kg	All values are in SI	E = mgh = 3 10 5 = 150 J Ep = mgh = 3 10 2 = 60 J Ek = E Ep = 150 60 = 90 J

Answer Ep = 60 J; Ek = 90 J.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IX. Simple Machines

33

Simple Machines

Simple machines transform force: they can change its magnitude and direction.
They include the lever, pulley, wheel and axle, inclined plane, wedge and screw.
A gain in force is usually accompanied by a loss in distance.

DEFINITIONS

Simple machines	devices used to transform force.
Lever	a rigid body able to rotate around a fixed support.
Lever arm	the shortest distance from the axis of rotation to a force s line of action.
Moment of a force	the product of a force and its lever arm; it characterizes turning effect.

FORMULAS AND RELATIONS

M = Fd	moment of force; d is lever arm; [M] = N m.
Fixed pulley: F = P	gives no force gain but changes the direction of force.
Movable pulley: F • P/2	ideally gives a force gain of about two.
Inclined plane: F • P h/l	ideal force along the plane; l is length and h is height.

VISUAL DIAGRAM

VISUAL DIAGRAM

Lever

Pulley

Inclined plane

Wedge

NOTE The golden rule of mechanics: no simple machine gives a gain in work; whatever is gained in force is lost in distance.

Chapter IX. Simple Machines

33 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 33

Problem. A load is lifted with a movable pulley by applying force 100 N. Ignore pulley weight and friction. Find the load weight.

Given	SI	Solution
F = 100 N P ?	All values are in SI	F = P/2 P = 2F P = 2 100 = 200 N

Answer P = 200 N.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IX. Simple Machines

34

Centre of Mass of Bodies

The centre of mass is the point that moves as the body would move if all of its mass were concentrated there.

For homogeneous symmetric bodies, it often coincides with the geometric centre, but it may lie outside the body.

Stability depends on the position of the centre of gravity and the support area.

DEFINITIONS

Centre of mass	a point whose motion characterizes the translational motion of a body.
Centre of gravity	the point of application of the resultant gravitational force on all parts of a body.
Stable equilibrium	equilibrium in which a body returns after a small displacement.
Unstable / neutral equilibrium	equilibrium in which a body moves farther away after a small displacement / remains in a new equilibrium position.

FORMULAS AND RELATIONS

Centre of gravity over support area stable	the vertical through the centre of gravity must pass through the support; otherwise the body tips over.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Use a plumb line to find the centre of gravity of a flat irregular shape: the intersection of vertical lines from two suspension points is the required point.

Chapter IX. Simple Machines

35

Condition for Lever Equilibrium

A lever is in equilibrium when the turning effects of forces balance each other.

A larger force can be balanced by a smaller force if the smaller force acts at a longer arm.

The principle of moments applies to any body rotating about a fixed axis.

DEFINITIONS

Lever equilibrium	a state in which a lever does not rotate.
Principle of moments	the sum of moments turning a body clockwise equals the sum of moments turning it counter-clockwise.
Equal-arm balance	a beam balance with equal arms; in equilibrium, the masses on its pans are equal.

FORMULAS AND RELATIONS

F /F = d /d	forces acting on a lever are inversely proportional to their arms.
F d = F d	equilibrium of two forces: their moments are equal.
Mclockwise = Mcounter-clockwise	principle of moments for several forces.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE A lever arm is always measured perpendicular to the force s line of action, not along the lever.

Chapter IX. Simple Machines

35 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 35

Problem. A 300 N force acts on the shorter arm of a lever and a 20 N force on the longer arm. The short arm is 5 cm. Find the longer arm.

Given	SI	Solution
F = 300 N F = 20 N d = 5 cm d ?	d = 0.05 m	F d = F d d = F d /F d = 300 0.05/20 d = 0.75 m

Answer d = 0.75 m = 75 cm.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter IX. Simple Machines

36

Efficiency (‡)

In real machines, part of the work is spent on friction, deformation and other losses.
Useful work is the intended work; total work is all work done by the applied force.
Efficiency shows what fraction of total work or power becomes useful.

DEFINITIONS

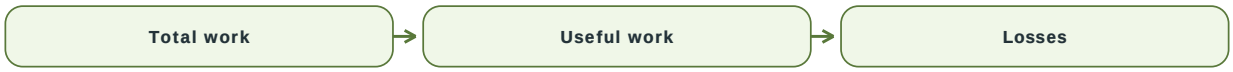
Useful work Auseful	the work required to achieve the purpose.
Total work Atotal	the work actually done by the applied force.
Efficiency ‡	the ratio of useful work or power to total work or power; it is usually expressed as a percentage.

FORMULAS AND RELATIONS

‡ = Auseful/Atotal 100%	efficiency calculated from work.
‡ = Nuseful/Ntotal 100%	efficiency calculated from power.
‡ < 100%	for real mechanisms because unavoidable energy losses occur.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Increase efficiency by reducing friction, using lubricants and bearings, and reducing unnecessary deformation and energy losses.

Chapter IX. Simple Machines

36 WORKED EXAMPLE

Worked Textbook Problem

Exercise in 36

Problem. A 20 kg load is pulled uniformly up a 2 m inclined plane by a force of 40 N. Its height is 10 cm. Find the efficiency.

Given	SI	Solution
m = 20 kg F = 40 N l = 2 m h = 10 cm g = 10 N/kg	h = 0.1 m	A = mgh = 20 10 0.1 = 20 J A = Fl = 40 2 = 80 J = A /A 100% = 20/80 100% = 25%

Answer = 25%.

Check the layout: write the quantities in SI choose the formula substitute values with units state the unit in the answer.

Chapter X. Space and Earth

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The Science of Celestial Bodies

Astronomy studies celestial bodies and their systems: their motion, structure, origin and development.

Historically, geocentric and heliocentric models of the world were proposed.

The observations and calculations of Copernicus, Galileo, Kepler and Newton formed the modern view of celestial motion.

DEFINITIONS

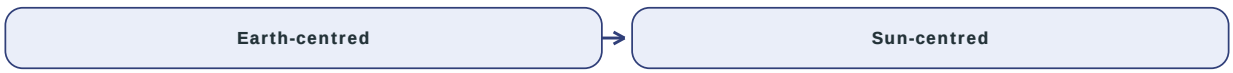
Astronomy	the science of motion, structure, origin and development of celestial bodies and their systems.
Geocentric system	a system with Earth at the centre.
Heliocentric system	a system with the Sun at the centre.
Celestial bodies	stars, planets, satellites, asteroids, comets and other space objects.

FORMULAS AND RELATIONS

Geocentrism: Earth at the centre	the model of Aristotle and Ptolemy; loops in planet motion were explained by epicycles.
Heliocentrism: Sun at the centre of the Solar System	the model of Copernicus; apparent loops result from different orbital speeds of planets.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE Galileo s telescopic observations supported heliocentric ideas; Kepler described planetary motion, and Newton explained it by gravitation.

Chapter X. Space and Earth

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The Solar System

The star Sun is at the centre of the Solar System, and eight planets orbit it.
Planets are divided into terrestrial planets and giant planets; most planets have satellites.
The system also contains asteroids, comets, meteoroids and belts of small bodies.

DEFINITIONS

Planet	a large celestial body orbiting the Sun; the school model includes eight planets.
Asteroid	a small Solar System body moving around the Sun in an orbit.
Comet	a small body on an elongated orbit that forms a gas-and-dust tail when near the Sun.
Meteor / meteorite	a meteor is light from a small body burning in the atmosphere; a meteorite is a part that reaches Earth s surface.

FORMULAS AND RELATIONS

Terrestrial planets	Mercury, Venus, Earth and Mars; they have solid surfaces.
Giant planets	Jupiter, Saturn, Uranus and Neptune; large, mainly gaseous planets.
1 AU • 150 000 000 km	astronomical unit: the average Earth Sun distance.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE The main asteroid belt lies between Mars and Jupiter; the Kuiper belt lies beyond Neptune. Mercury and Venus have no natural satellites.

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Fundamentals of the Calendar

A calendar is a system for counting long periods based on recurring celestial motions.

The main natural time units are the solar day, lunar month and solar (tropical) year.

Seasons are caused by the tilt of Earth s axis and its revolution around the Sun, not by changing Earth Sun distance.

DEFINITIONS

Calendar	a system for counting long periods of time based on the periodic motion of celestial bodies.
Tropical year	the time for one complete revolution of Earth around the Sun: about 365 days 5 h 48 min 46 s.
Solar calendar	a calendar that reconciles days with the solar year.
Equinox and solstice	days related to Earth s orbital position: at an equinox day and night are equal; at solstice day length reaches a maximum or minimum.

FORMULAS AND RELATIONS

Julian calendar: average year 365.25 days	every fourth year is a leap year; the calendar slowly drifts from the tropical year.
Gregorian rule	a year divisible by 4 is leap; century years are not leap unless divisible by 400.
Earth s axial tilt • 23ˆ27′	the cause of different Sun heights, seasons and different daylight length by latitude.

VISUAL DIAGRAM

VISUAL DIAGRAM



NOTE In the Northern Hemisphere, the summer solstice is about 22 June and the winter solstice about 22 December; at equinoxes day and night are approximately 12 hours each.